More Novel Data Streams

Mobility, weather, and news, borrowing what the field already uses

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A different question

The streams so far (search, Wikipedia, wastewater) mostly answer **“how much disease is here now?”** These three answer different questions:

- **Mobility**: where is it *coming from*?
- **Weather**: what is *driving* it?
- **News**: what is *emerging* that has no data yet?

Honest framing

These are streams other groups pioneered: **GLEAM** (Vespignani’s lab) for mobility, **Shaman et al.** for weather, **HealthMap / ProMED** for news. We credit them, and we show how to reach *free, scrapable* versions with an agent.

Roadmap

Human mobility

Weather

News alerts

Social and survey signals

Pulling it together

Human mobility

Disease travels with people, not with distance

Epidemics spread along how people actually move. Big, well-connected places seed smaller ones (the gravity / metapopulation intuition behind flu spread).

- **Ground** movement (commuting) drives local, radial spread between nearby counties.
- **Air** travel drives long-range importation, seeding distant places.

So two questions: which neighbors will seed me next, and which far-off places import risk to me?

The gold standard: GLEAM (Vespignani / MOBS lab)

The most established mobility-driven epidemic model, and worth knowing by name.

- **Air travel:** the **OAG** and **IATA** databases, roughly 3,800 airports and 4M+ connections.
- **Ground:** a **commuting** database from 40+ national statistics offices, 78,000+ regions, 5M+ connections, mapped with Voronoi cells around airports.
- **Population:** WorldPop / Gridded Population of the World.

The catch for a hands-on class

OAG, IATA, and SafeGraph are **licensed**. We cannot scrape them live. So we use **free proxies** and keep cached snapshots of the licensed ones.

Meet the source: US Census commuting data (LODES)

Every year the Census links **where people live** to **where they work**, from employment records. It is free, it covers the whole United States, and it goes down to the neighbourhood.

- Think of it as: *"how many people sleep in county A and work in county B?"*
- That is the commuting shed, and it is the ground truth for which neighbours you actually exchange people with every single day.

Open and demo
`onthemap.ces.census.gov`
A clickable map: pick a county, and watch the arrows showing where its workers come from. Raw files live at `lehd.ces.census.gov/data/`.

Ground mobility you can actually scrape

- **US Census LODES** (origin-destination commuting) is free and detailed: who lives where and works where, down to the census tract.
- **Google COVID-19 Community Mobility** (archived 2020–2022) and the **ODT Flow Explorer** are free downloads; **SafeGraph** is licensed, so cached.

Prompt

```
Using US Census LODES origin-destination data, for Fulton
County, GA, rank the counties that send the most commuters
in. Return the top 10 as a tidy table. These are the
neighbors whose outbreaks tend to reach us first.
```

Meet the source: OpenFlights

A volunteer-built, openly published database of the world's **airports and airline routes**. Not passenger counts, but the *wiring diagram* of global air travel.

- Think of it as: *"which airports have a direct flight to mine, and where are they?"*
- That wiring is what turns a distant outbreak into a local case, which is the whole idea behind importation risk.

Open and demo
`openflights.org`

Search an airport and it draws every route on a world map. The underlying `routes.dat` and `airports.dat` are plain text files we read directly.

Air mobility you can actually scrape

- **OpenFlights** publishes a free, no-key table of airline routes: the *structure* of air connectivity, a workable stand-in for OAG/IATA.
- **OpenSky Network** has real flight data (ADS-B), but its API now needs a free account and OAuth2 credentials (anonymous requests get a 403), so it is a take-home rather than an in-class pull.
- Cross-check which countries a place **exchanges** with using Wikipedia (a trick from our own planning), then reason about importation risk.

Prompt

```
Rank the countries that connect to Atlanta (ATL) as a
proxy for air-importation risk. Use the free OpenFlights
routes.dat + airports.dat (no key): count inbound routes
to ATL by origin airport, map each to its country, and
plot the top 10 excluding domestic.
```

Asked for: Meta's AI for Good datasets

A participant asked whether we could work with Meta's AI for Good data. Mostly **yes**, because Meta publishes through the **Humanitarian Data Exchange** (HDX), which has a free, open API.

Dataset	**In class?**	**How**
Relative Wealth Index	**yes**	free per-country CSVs on HDX
Movement Distribution Maps	**yes**	free on HDX, CC BY, current to June 2026
Activity Space Maps	not directly	not on HDX; Meta's own portal
Points of Interest	not directly	not on HDX under Meta

Two of the four are usable today, with no application. Details next slide.

Meet the source: Meta Data for Good (via HDX)

Meta turns its data into **aggregated, privacy-protected** public goods and gives them away through the humanitarian data portal the UN and NGOs use.

- **Relative Wealth Index**: how well-off each 2.4 km square is. A useful *confounder*: wealth shapes housing, water storage and care-seeking.
- **Movement Distribution**: how far people travel from home, by region and day. The behaviour channel disease travels through.
- **224 datasets** in total, including Social Connectedness and population density maps.

Open and demo

`data.humdata.org/organization/meta` lists all 224, each with a Download button. Meta's own write-ups: `dataforgood.facebook.com`.

Weather

Weather as a modulator, not a case count

Weather does not count cases; it **shifts the conditions** for transmission.

- **Influenza: absolute humidity** is the dominant driver. Shaman et al. showed virus survival and transmission fall as absolute humidity rises, and built humidity-driven flu forecasts.
- **Vector-borne (dengue, Zika): temperature and rainfall** govern the mosquito life cycle, so they modulate outbreak timing and size.

Mauricio picks up the modeling of weather-as-modulator this afternoon; here we just get the data.

Meet the source: Open-Meteo

A free weather service built for programs rather than people: give it a latitude, longitude and date range, and it returns decades of daily weather. **No account, no key, no cost** for our use.

- Think of it as: *"the weather history of any point on Earth, as a table."*
- Behind it sits **ERA5**, the research-grade reanalysis that most humidity-and-flu papers use, at `cds.climate.copernicus.eu`.

Open and demo

`open-meteo.com/en/docs/historical-weather-api`

A live form: type a city, tick *temperature* and *dew point*, and it builds the exact web address that returns the data. Paste that link in a browser to see the numbers.

Weather you can actually scrape

- **Open-Meteo** gives free historical and forecast weather with **no key**: ideal for an agent.
- **ERA5** (Copernicus) is the research-grade reanalysis (temperature and dew point at 0.25 degrees) behind most humidity studies.

Prompt

```
Using the Open-Meteo historical API (no key), pull daily
temperature and dew point for Atlanta since 2016. Compute
weekly absolute humidity from them, and overlay it on flu
ED visits so I can see whether low humidity precedes flu
season.
```

News alerts

News: catching what has no data yet

For a genuinely **emerging** outbreak, there is no search history and no wastewater site. The earliest signal is a report.

- **ProMED** (human-curated, since 1994) and **HealthMap** (news wires, RSS, ProMED, WHO, auto-parsed and geocoded) are the established event-based systems. ProMED flagged COVID roughly a day before official notification.
- These feed WHO's EIOS and are used by CDC and WHO.

This is also a stream in Mauricio and Kogan's multi-source COVID forecasting: searches + news alerts + GLEAM.

Meet the sources: HealthMap, ProMED, GDELT

Three ways the world finds out about an outbreak before the data exists.

- **HealthMap** reads news and official reports automatically and pins them on a world map. `healthmap.org`
- **ProMED** is the opposite: real human experts, posting curated outbreak reports since 1994. It flagged COVID about a day before the official notice. `promedmail.org`
- **GDELT** indexes world news in 100+ languages and gives it away through a free API, which is why it is the one we can scrape in class. `gdeltproject.org`

Open and demo

Open `healthmap.org` and zoom to a region: every dot is a report. Then open `promedmail.org` and read one post, to feel the difference between an automated map and a curated expert alert.

News you can actually scrape

GDELT monitors world news in 100+ languages, is 100% free, and has an API, which makes it perfect for an agent. Or point the agent at a news filter directly.

Prompt

```
Using the GDELT DOC 2.0 API, find news articles from the
past month that mention dengue outbreaks. Return date,
country, and headline; then summarize which countries are
newly reporting activity, as an emerging-outbreak watch.
```

Noisy by nature: news reflects attention and reporting, so a human read is part of the loop.

Social and survey signals

What happened to Twitter and Facebook

The streams that powered 2020 research have mostly closed their doors.

Source	**Today**	**Why it matters**
X / Twitter posts	paid API since 2023	the classic firehose is gone
Facebook posts	CrowdTangle shut 2024	research access by application only
Bluesky	**open**	the workable social option
Facebook symptom survey	**open** via CMU Delphi	the good part survived

The lesson worth remembering

Our own COVID exercise file still has **Twitter, Kinsa and Cuebiq** columns, and all three of those streams are now closed or gone. Novel data streams are **fragile**; that is the argument for an agent that can re-scrape whatever exists today.

Meet the source: Bluesky

A public social network with an **open door**: anyone can read public posts without an account, which is exactly what Twitter stopped allowing.

- Think of it as: *"a Twitter-shaped network you are still allowed to read."*
- One catch: **keyword search** needs a login, but **finding accounts** and **reading their public posts** does not. So we watch *accounts* (epidemiologists, health agencies) rather than searching words.

Open and demo
`bsky.app`
Search *dengue* or *outbreak* in the website's own search box (the browser can do what the open API will not), then open a public-health account and scroll its feed. That feed is exactly what our notebook reads.

Meet the source: CMU Delphi (and the Facebook survey)

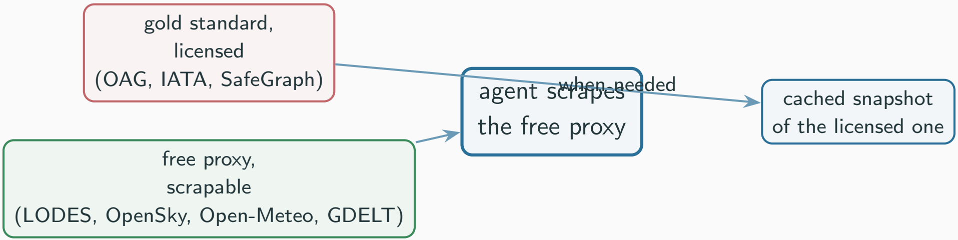
You cannot get Facebook *posts*, but you can get something better for our purposes: for two years Facebook asked millions of users **"are you or anyone in your household ill?"**, and Carnegie Mellon turned it into a public health signal.

- Think of it as: *"a nationwide symptom survey, every day, free."*
- It even ships **error bars and sample sizes**, which almost no other novel stream does. It ran 2020 to June 2022, so it is a benchmark, not a live feed.
- Delphi is a hub: medical claims, hospital admissions and more sit behind the same free API.

Open and demo
`delphi.cmu.edu/covidcast/survey-results`
Delphi's **CTIS Dashboard**, their page for what they themselves label "*the Facebook Covid Survey*". Background at `delphi.cmu.edu/epidemic-signals/ctis`; the exact signal we pull (`smoothed_wcli`) is defined in the signal dictionary at `cmu-delphi.github.io/delphi-epidata`

Pulling it together

The pattern behind all three



- Same five-step loop, same verify habit as every other stream.
- These are **context** signals (spread, drivers, emergence), noisier than a local case proxy. Treat them as complements, and validate.

Where this connects

- **This afternoon:** the network / ARGONet idea borrows signal from neighbors, which is exactly what **ground mobility** tells you to do. Mauricio also covers **weather** and **news** for emerging outbreaks.
- **Day 3 capstone:** pick the streams that fit your question. Importation risk? Use air mobility. Seasonality? Weather. Emerging threat? News.

You now know the field's established streams by name, and a free, agent-scrapable way into each.

Credits and sources

- **Mobility / GLEAM:** Balcan et al., *Multiscale mobility networks and the spatial spreading of infectious diseases*, PNAS 2009; the GLEAM Project (MOBS lab, Northeastern; Vespignani). Data: OAG, IATA, national commuting statistics, WorldPop.
- **Free mobility proxies:** US Census LODES; OpenSky Network; Google COVID-19 Community Mobility Reports (archived); ODT Flow Explorer.
- **Weather:** Shaman & Kohn (PNAS 2009) and Shaman et al. (PLoS Biology 2010; PNAS 2012) on absolute humidity and influenza. Data: Open-Meteo; ERA5 (Copernicus).
- **News:** ProMED (ISID); HealthMap (Brownstein / Freifeld); WHO EIOS; GDELT. Multi-source forecasting: Kogan, Santillana et al., *Science Advances* 2021.

Questions?

Repo: `github.com/Shihao-Yang/sismid2026`